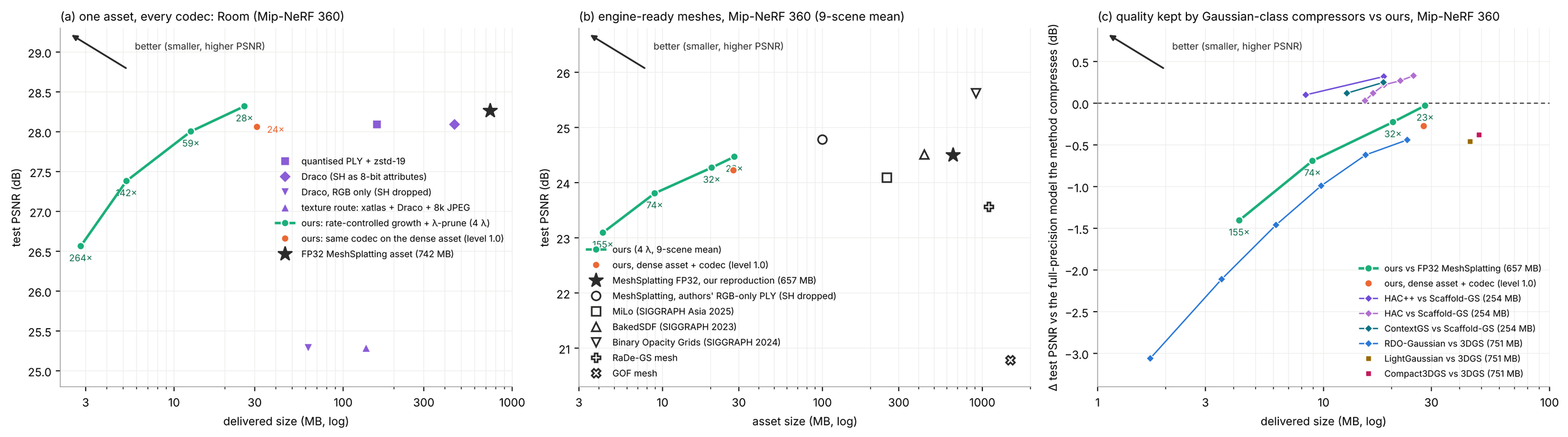
(a) one asset, every codec: Room (Mip-NeRF 360)
better (smaller, higher PSNR)
test PSNR (dB)
delivered size (MB, log)
264×
142×
59×
28×
24×
quantised PLY + zstd-19
Draco (SH as 8-bit attributes)
Draco, RGB only (SH dropped)
texture route: xatlas + Draco + 8k JPEG
ours: rate-controlled growth + λ-prune (4 λ)
ours: same codec on the dense asset (level 1.0)
FP32 MeshSplatting asset (742 MB)
(b) engine-ready meshes, Mip-NeRF 360 (9-scene mean)
better (smaller, higher PSNR)
test PSNR (dB)
asset size (MB, log)
155×
74×
32×
ours (4 λ, 9-scene mean)
ours, dense asset + codec (level 1.0)
MeshSplatting FP32, our reproduction (657 MB)
MeshSplatting, authors' RGB-only PLY (SH dropped)
MiLo (SIGGRAPH Asia 2025)
BakedSDF (SIGGRAPH 2023)
Binary Opacity Grids (SIGGRAPH 2024)
RaDe-GS mesh
GOF mesh
(c) quality kept by Gaussian-class compressors vs ours, Mip-NeRF 360
better (smaller, higher PSNR)
Δ test PSNR vs the full-precision model the method compresses (dB)
delivered size (MB, log)
155×
74×
32×
23×
ours vs FP32 MeshSplatting (657 MB)
ours, dense asset + codec (level 1.0)
HAC++ vs Scaffold-GS (254 MB)
HAC vs Scaffold-GS (254 MB)
ContextGS vs Scaffold-GS (254 MB)
RDO-Gaussian vs 3DGS (751 MB)
LightGaussian vs 3DGS (751 MB)
Compact3DGS vs 3DGS (751 MB)